

3 Chemical bonding

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Ionic bonding is electrostatic attraction between oppositely charged ions. Covalent bonds share electron pairs; electronegativity difference gives polarity, but molecular shape determines overall dipole. VSEPR predicts electron-pair arrangement; lone pairs repel more strongly than bonding pairs. Hydrogen bonding requires H bonded to N, O or F and a suitable lone pair.

Worked example: follow the reasoning

NH_3 has four electron regions around N; one lone pair gives trigonal pyramidal shape and about 107° bond angle.

Quick check · multiple choice

A Why is water polar? [1]

A It is linear

B O–H bonds are nonpolar

C Bent shape prevents dipole cancellation

D It has no lone pairs

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why water has a bent shape and a permanent dipole. [3]

Answers and explanation

Quick check: C. Bent shape prevents dipole cancellation

Question 1 [3 marks]

O has two bonding and two lone pairs [1]; lone-pair repulsion produces bent arrangement [1]; O-H bond dipoles do not cancel [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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